

# Probabilistic Machine Learning

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## Lecture 04: Naive Bayes and the Generative Route

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# Where We Left Off

Lecture 03 took two shortcuts and named what each drops:

| Shortcut           | Instead of the ideal                               | What is dropped                          |
|--------------------|----------------------------------------------------|------------------------------------------|
| Keep the peak      | the most probable parameter (MLE, MAP)             | the spread: how sure we are              |
| Only what you need | model $p(y \mid x)$ directly — logistic regression | $p(x)$ : every question about the inputs |

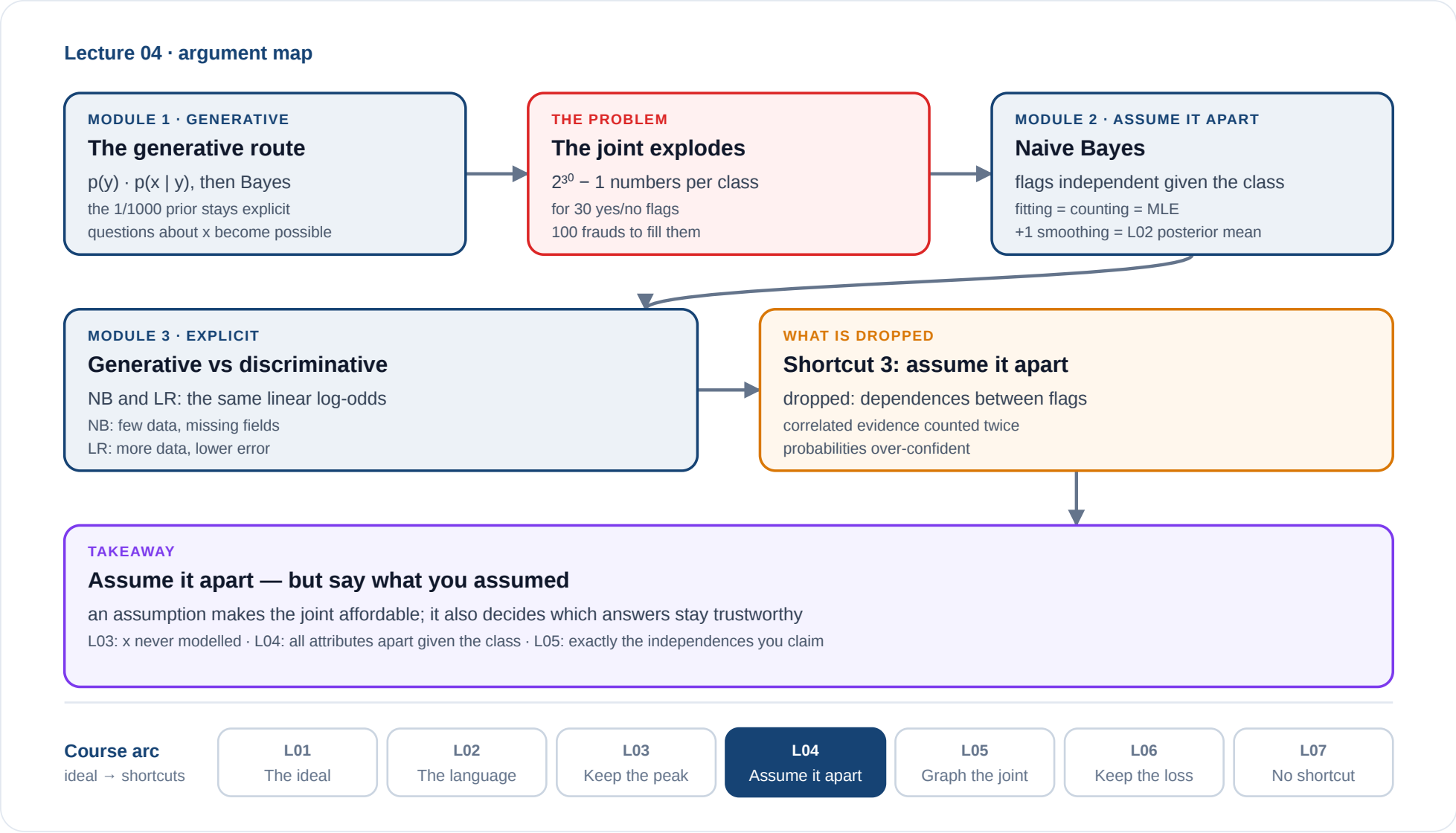
Today the inputs come back. The **generative route** models how the data arises, class by class, and converts with Bayes:

$$p(x, y) = p(y) \cdot p(x \mid y)$$

### THE CATCH

$p(x \mid y)$  is a joint over every attribute — and it explodes. Paying for it is **shortcut 3: assume it apart.**

# Argument Map



# The Generative Route

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**Guiding Question:** Logistic regression never asked what a fraudulent transaction *looks like*. What would it take to model that — and why does nobody do it in full?

# Two Routes to the Same Posterior

A labelled table of transactions supports **both** routes to  $P(\text{fraud} \mid x)$ , where  $x$  is the vector of red flags:

## DISCRIMINATIVE — LECTURE 03

### Model $p(y \mid x)$ directly

Logistic regression: a score for each  $x$ .  
Nothing is said about how  $x$  arises.

## GENERATIVE — TODAY

### Model $p(y)$ and $p(x \mid y)$

How does a *fraudulent* transaction look, and how an honest one? Then convert with Bayes.

$$P(y \mid x) = \frac{p(x \mid y) \cdot P(y)}{\sum_{y'} p(x \mid y') P(y')}$$

# Why Model How Fraud Looks?

Logistic regression already gives  $P(\text{fraud} \mid x)$ . What can a model of  $p(x \mid y)$  do that it cannot?

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Logistic regression already gives  $P(\text{fraud} \mid x)$ . What can a model of  $p(x \mid y)$  do that it cannot?

- ✓ **Keep the prior explicit.** The 1 in 1,000 of Lecture 02 is a separate factor: a new market with 1 in 200 changes  $P(y)$ , not the whole model.
- ✓ **Handle a missing field.** An attribute nobody observed can be summed out of  $p(x \mid y)$ .
- ✓ **Notice the strange.** A transaction with tiny  $p(x)$  under *both* classes is unlike anything seen — a score alone cannot say so.

The price:  $p(x \mid y)$  is a **joint distribution over all attributes**, one per class.

# The Joint Explodes

**30**

yes/no red flags per transaction



**$2^{30} - 1$**

≈ 1.07 billion numbers for  $p(x \mid \text{fraud})$  alone



**100**

frauds among 100,000 transactions to estimate them

Almost every cell of the table is **empty**. Its count is 0, so its maximum likelihood estimate is 0 — every fraud that looks slightly new becomes "impossible".

# Mini-Exercise

A fraud model uses only **three** yes/no flags.

How many numbers does the generative model need — both classes and the class prior together?

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**15.** Each class needs a full table over  $2^3 = 8$  flag patterns: 7 free numbers (they sum to 1). Two classes give 14, and  $P(\text{fraud})$  adds one.

# Takeaway

The generative route models **how the data arises** —  $p(y)$  and  $p(x | y)$  — and converts with Bayes. It keeps the prior explicit and can answer questions about  $x$  itself, but it must pay for a **joint over all attributes**, which explodes.

# Naive Bayes: Assume It Apart

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**Guiding Question:** Its central assumption is almost always false, and it classifies well anyway. How can both be true?

# Conditional Independence

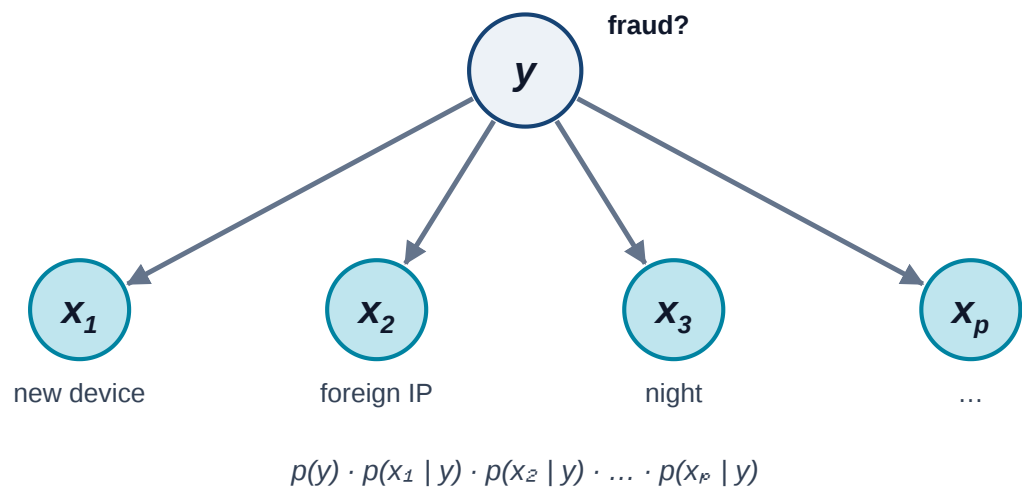
**Definition.**  $A$  and  $B$  are **conditionally independent given  $C$** , written  $A \perp\!\!\!\perp B \mid C$ , iff

$$P(A \cap B \mid C) = P(A \mid C) \cdot P(B \mid C) \iff P(A \mid B \cap C) = P(A \mid C)$$

Reading: **once  $C$  is known,  $B$  tells us nothing more about  $A$ .**

Among frauds only: does a new device make a foreign IP more likely?

# The Naive Assumption



Assume the attributes are **independent given the class**:

$$p(x_1, \dots, x_p \mid y) = \prod_{j=1}^p p(x_j \mid y)$$

| flags | full joint, per class | naive |
|-------|-----------------------|-------|
| 3     | 7                     | 3     |
| 30    | 1,073,741,823         | 30    |

**Shortcut 3 — assume it apart.** Dropped: every dependence *within* a class.

# Fitting Is Counting

$$\hat{P}(y) = \frac{n(y)}{n} \quad \hat{P}(x_j = v \mid y) = \frac{n(x_j = v, y)}{n(y)}$$

## Lecture 02

Each attribute, within one class, is a **categorical** variable — for a yes/no flag, a Bernoulli.

## Lecture 03

Its maximum likelihood estimate is the **relative frequency** — the same derivation as  $h/n$  for the coin.

Naive Bayes is fitted in one pass over the table, with no optimiser — and every number it uses is a **peak**. Shortcut 1 is taken as well.

# The Classifier

$$P(y \mid x_1, \dots, x_p) = \frac{P(y) \prod_j p(x_j \mid y)}{\sum_{y'} P(y') \prod_j p(x_j \mid y')}$$

For two classes, divide the two posteriors — the normaliser cancels:

$$\underbrace{\frac{P(\text{fraud} \mid x)}{P(\text{honest} \mid x)}}_{\text{posterior odds}} = \underbrace{\frac{P(\text{fraud})}{P(\text{honest})}}_{\text{prior odds}} \cdot \prod_{j=1}^p \underbrace{\frac{p(x_j \mid \text{fraud})}{p(x_j \mid \text{honest})}}_{\text{likelihood ratio LR}_j}$$

Every observed attribute **multiplies the odds** by its likelihood ratio: above 1 it speaks for fraud, below 1 against.

# Three Red Flags

100,000 transactions, 100 of them fraud: prior odds 1 : 999.

| flag       | among frauds | among honest | likelihood ratio |
|------------|--------------|--------------|------------------|
| new device | 60%          | 10%          | 6                |
| foreign IP | 70%          | 5%           | 14               |
| night      | 50%          | 20%          | 2.5              |

A transaction shows all three flags. Is it more likely fraud than not?

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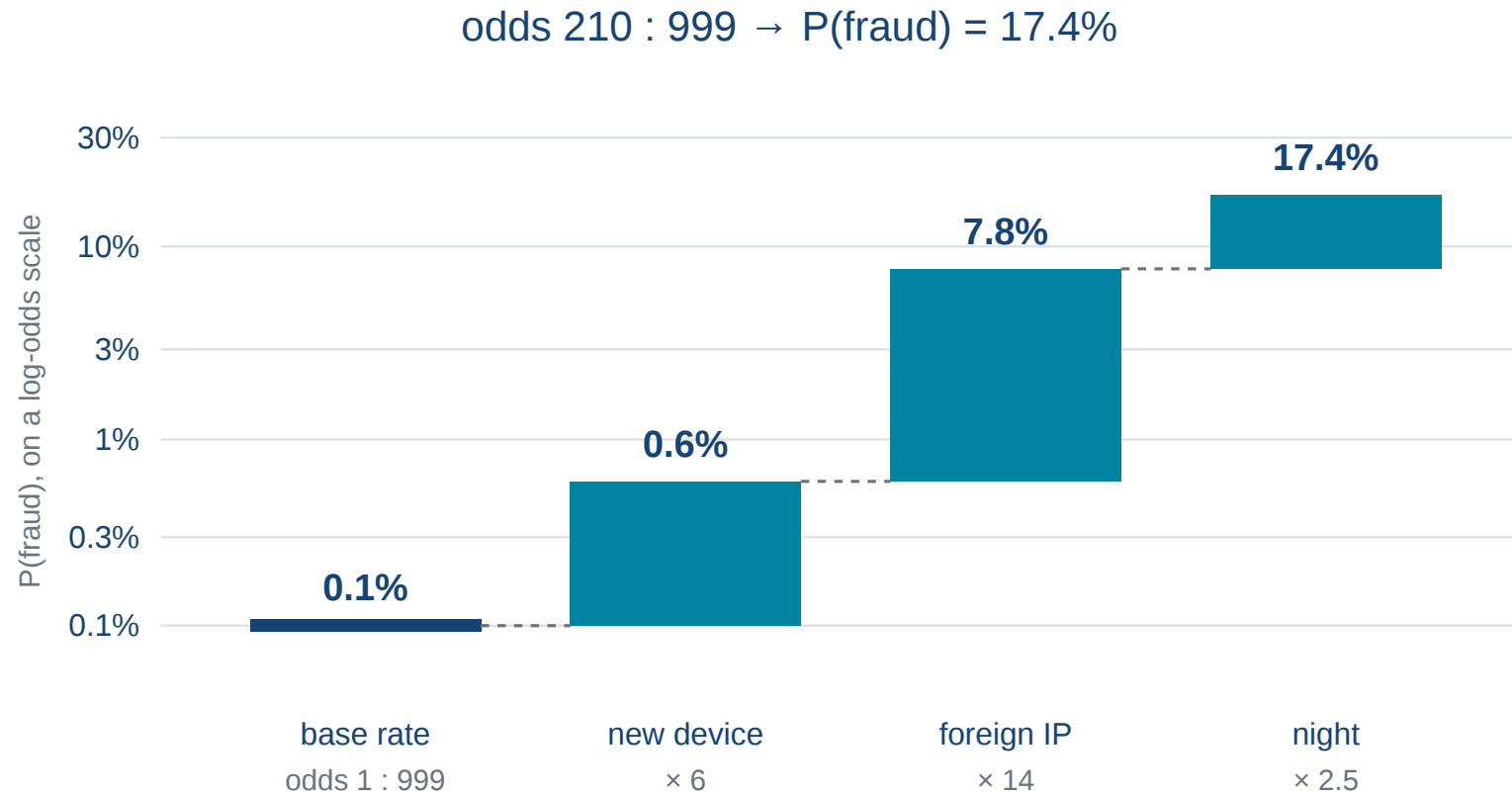
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A transaction shows all three flags. Is it more likely fraud than not?

$$\text{odds} = \frac{1}{999} \cdot 6 \cdot 14 \cdot 2.5 = \frac{210}{999} \implies P(\text{fraud} \mid x) = \frac{210}{1,209} \approx 17\%$$

Three red flags, each one strong — and still **mostly honest**. The base rate of Lecture 02, once more.

# Evidence Adds Up



**Predict before each click:** which flag moves the probability most?

# One Zero Vetoes Everything

A new flag: **payment to a crypto exchange**. Among the 100 frauds it never occurred.

$$\hat{P}(\text{crypto} \mid \text{fraud}) = \frac{0}{100} = 0$$

A transaction with this flag gets posterior odds **exactly 0** — whatever the other thirty flags say. One unseen value vetoes all other evidence.

## KEEP THE PEAK, FAILS

"Not seen in 100 frauds" became "impossible for a fraud" — the three heads in three flips of Lecture 03, in a single cell.

# Smoothing Is a Prior

**Laplace smoothing** adds one pseudo-count to every value of attribute  $j$  ( $k_j$  values):

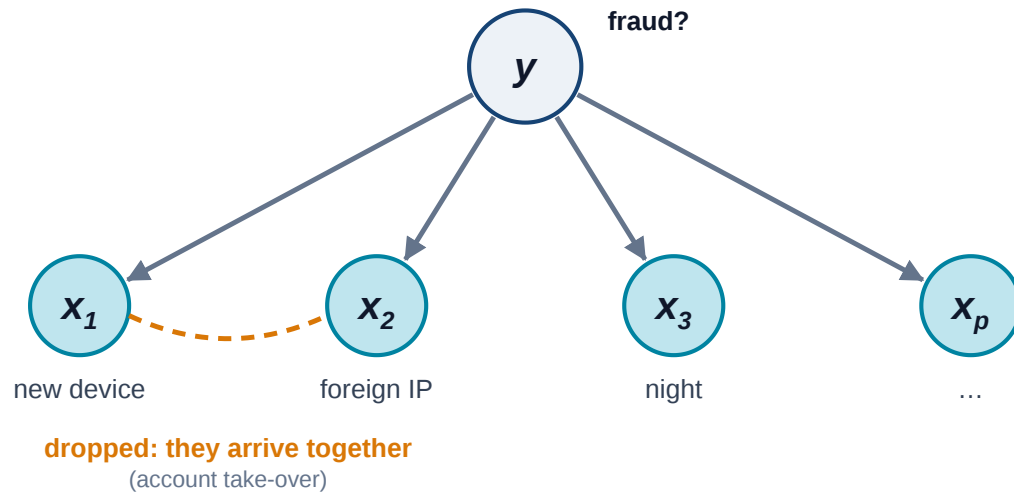
$$\hat{P}(x_j = v \mid y) = \frac{n(x_j = v, y) + 1}{n(y) + k_j}$$

For a yes/no flag ( $k_j = 2$ ) this is  $\frac{h+1}{n+2}$  — the **posterior mean**  $\frac{a+h}{a+b+n}$  of Lecture 02 under the flat prior Beta(1, 1). For  $k_j > 2$  values, the flat Dirichlet does the same.

| count among 100 frauds | MLE (peak) | smoothed (mean)        |
|------------------------|------------|------------------------|
| 0                      | 0          | $1/102 \approx 0.010$  |
| 60                     | 0.60       | $61/102 \approx 0.598$ |

Under the flat prior the peak of 0 of 100 is still 0. Smoothing quietly **leaves the peak for the mean** — exactly where the peak was wrong.

# Counting Evidence Twice



In an account take-over, a new device and a foreign IP **arrive together**. Suppose both appear on 55% of frauds and 4% of honest transactions:

|               | ratio for the pair     | $P(\text{fraud})$ with night |
|---------------|------------------------|------------------------------|
| naive Bayes   | $6 \cdot 14 = 84$      | 17.4%                        |
| the real pair | $0.55/0.04 \approx 14$ | 3.3%                         |

Correlated flags are **counted twice**, so naive Bayes is **over-confident**. Its ranking of transactions often survives; its probabilities do not.

# Mini-Exercise

A bug feeds the **foreign-IP** flag into the model twice.

Does the fraud probability of our three-flag transaction go up, down, or stay — and what does that say about correlated attributes?

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**Up — from 17% to about 75%.** The odds get a second factor 14: 2,940 : 999. Naive Bayes cannot tell a copy from new evidence; a strongly correlated attribute is almost a copy.

# Takeaway

Naive Bayes **assumes the joint apart** — attributes independent given the class — and **keeps the peak** of every count; smoothed, it keeps the Beta posterior *mean* instead. It is cheap, needs little data and ranks well. Its probabilities are over-confident wherever attributes move together.

# Generative vs Discriminative

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**Guiding Question:** Naive Bayes and logistic regression see the same fraud table. Which would you trust with 50 labelled frauds — and which with 50,000?

# The Same Line, Fitted Twice

Take the logarithm of the naive-Bayes odds. For yes/no flags  $x_j \in \{0, 1\}$  with  $\theta_{jy} = P(x_j = 1 \mid y)$ :

$$\log \frac{P(\text{fraud} \mid x)}{P(\text{honest} \mid x)} = w_0 + \sum_{j=1}^p w_j x_j, \quad w_j = \log \frac{\theta_{j1}}{\theta_{j0}} - \log \frac{1 - \theta_{j1}}{1 - \theta_{j0}}$$

Where have you seen a log-odds that is linear in  $x$ ?

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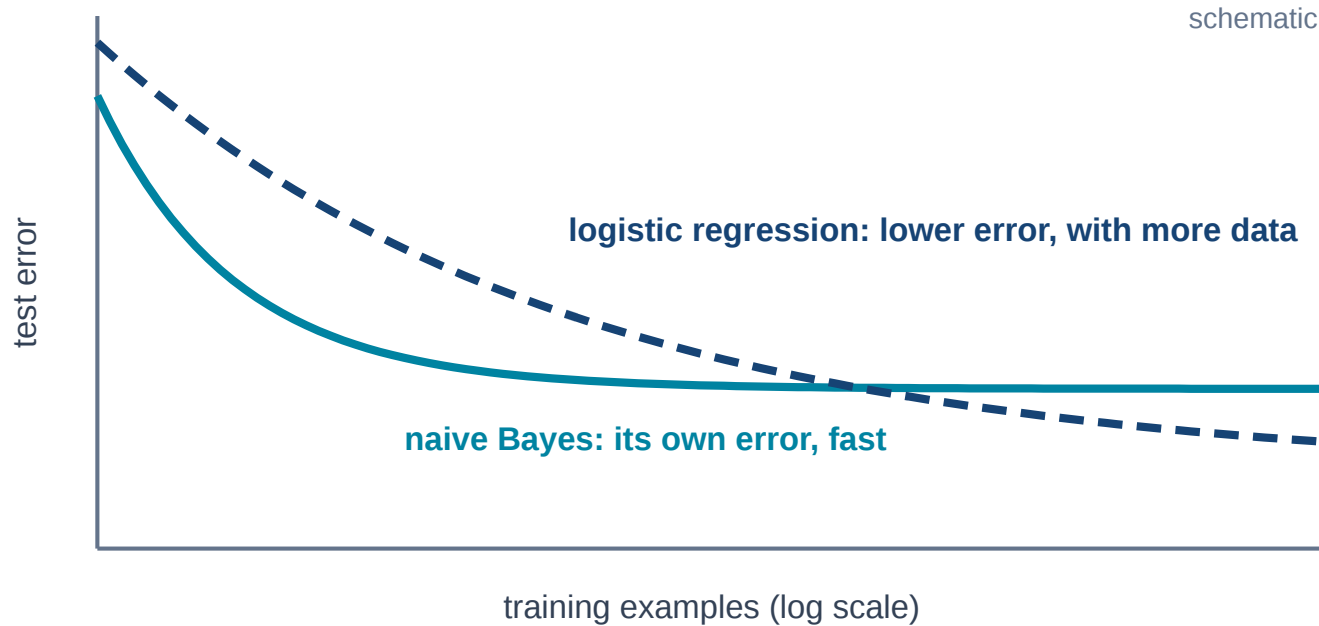
Where have you seen a log-odds that is linear in  $x$ ?

**Logistic regression** (Lecture 03):  $P(\text{fraud} \mid x) = \sigma(w_0 + w^\top x)$ . Same family of classifiers — naive Bayes sets the weights by **counting**, one attribute at a time; logistic regression by **maximising**  $\sum_i \log p(y_i \mid x_i)$ , all weights jointly.

# Two Ways to a Classifier

|                  | naive Bayes — generative       | logistic regression — discriminative            |
|------------------|--------------------------------|-------------------------------------------------|
| models           | $P(y)$ and $p(x   y)$          | $P(y   x)$ only                                 |
| fitted by        | counting (joint likelihood)    | iterative optimisation (conditional likelihood) |
| correlated flags | counted twice                  | credit shared                                   |
| a missing field  | drop its factor                | needs a value                                   |
| an unusual $x$   | low $p(x)$ : can be flagged    | still a confident score                         |
| a new base rate  | change $P(y)$                  | refit or correct the intercept                  |
| drops            | dependences between attributes | everything about $x$                            |

# Few Data, Many Data



**Few labelled frauds:** naive Bayes. Its assumption acts like a strong prior — little to estimate, so it settles fast.

**Many labelled frauds:** logistic regression. It pays for fewer assumptions with data, and ends lower.

Shown empirically and in theory by Ng & Jordan (2002). A wrong assumption costs accuracy in the limit and buys it early — the prior of Lecture 02, again.

# Mini-Exercise

Our three-flag transaction arrives **without the IP field** — the logging failed.

What does naive Bayes do with it — and what does logistic regression do?

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Our three-flag transaction arrives **without the IP field** — the logging failed.

What does naive Bayes do with it — and what does logistic regression do?

**Naive Bayes** drops the factor:  $\sum_v p(x_{\text{IP}} = v \mid y) = 1$ , so the odds become  $6 \cdot 2.5 : 999$  — about **1.5%**. **Logistic regression** has no model of  $x$ : it needs *some* value for  $x_{\text{IP}}$ , so someone must impute one, or train a second model without the field.

# Takeaway

Naive Bayes and logistic regression draw **the same kind of line**; they differ in what they model. The generative model pays with assumptions about  $x$  and gets answers about  $x$ ; the discriminative one pays with data.

| lecture | shortcut           | what is dropped                     |
|---------|--------------------|-------------------------------------|
| 03      | keep the peak      | how sure we are                     |
| 03      | only $p(y \mid x)$ | every question about $x$            |
| 04      | assume it apart    | every dependence between attributes |

Naive Bayes is a graph with one hub. **Lecture 05** draws the others — and keeps exactly the dependences we claim.